# Sea Ice Deep Fusion

## End-to-End Workflow

Satellite imagery + ICESat-2 laser altimetry fusion
for Arctic sea ice classification

| Best mIoU | Pixel Accuracy | Macro F1 |
|---|---|---|
| **0.8982** | **95.3 %** | **0.9453** |

**1** Thick Ice

**2** Thin Ice

**3** Water

3-class semantic segmentation · Tile-grouped train/test split · Focal loss

# Problem Overview & Data Sources



### Why is this hard?

- Thin ice is visually similar to thick ice in RGB
- Melt ponds create ambiguous color signatures
- Spatial resolution limits single-sensor accuracy
- Class imbalance: thin ice is rare but critical
- Generalization: test tile is geographically separate

### Our Solution: Multimodal Fusion

- Satellite RGB patches → spatial texture (U-Net)
- ICESat-2 ATL03 segments → height/roughness (LSTM)
- Deep fusion: combine features before final decision
- Transfer learning: pretrained ResNet-18 + sweep winner
- Squeeze-Excitation gating for adaptive channel weighting

### Satellite Images

Format: 128×128 RGB patches

Tiles: T02CNA, T02CNC (train)

T03CWT (test)

Pre-proc: ImageNet normalise

Augment: flip + rotate (train)

### ICESat-2 / ATL03

Format: 10 m along-track segments

Features (8 per segment):

h_cor_mean, h_diff

rel_height_min_elev, height_sd

pcnth/pcnt/bcnt/brate_mean

Window: 5 consecutive segments

### Segmentation Masks

3 classes encoded as RGB:

Red (255,0,0) → Class 0

= Thick Ice

Blue (0,0,255) → Class 1

= Thin Ice / Melt pond

Green (0,255,0) → Class 2

= Open Water

### Dataset Split Strategy (tile-grouped — no geographic leakage)

**Train (90 %)**
T02CNA + T02CNC

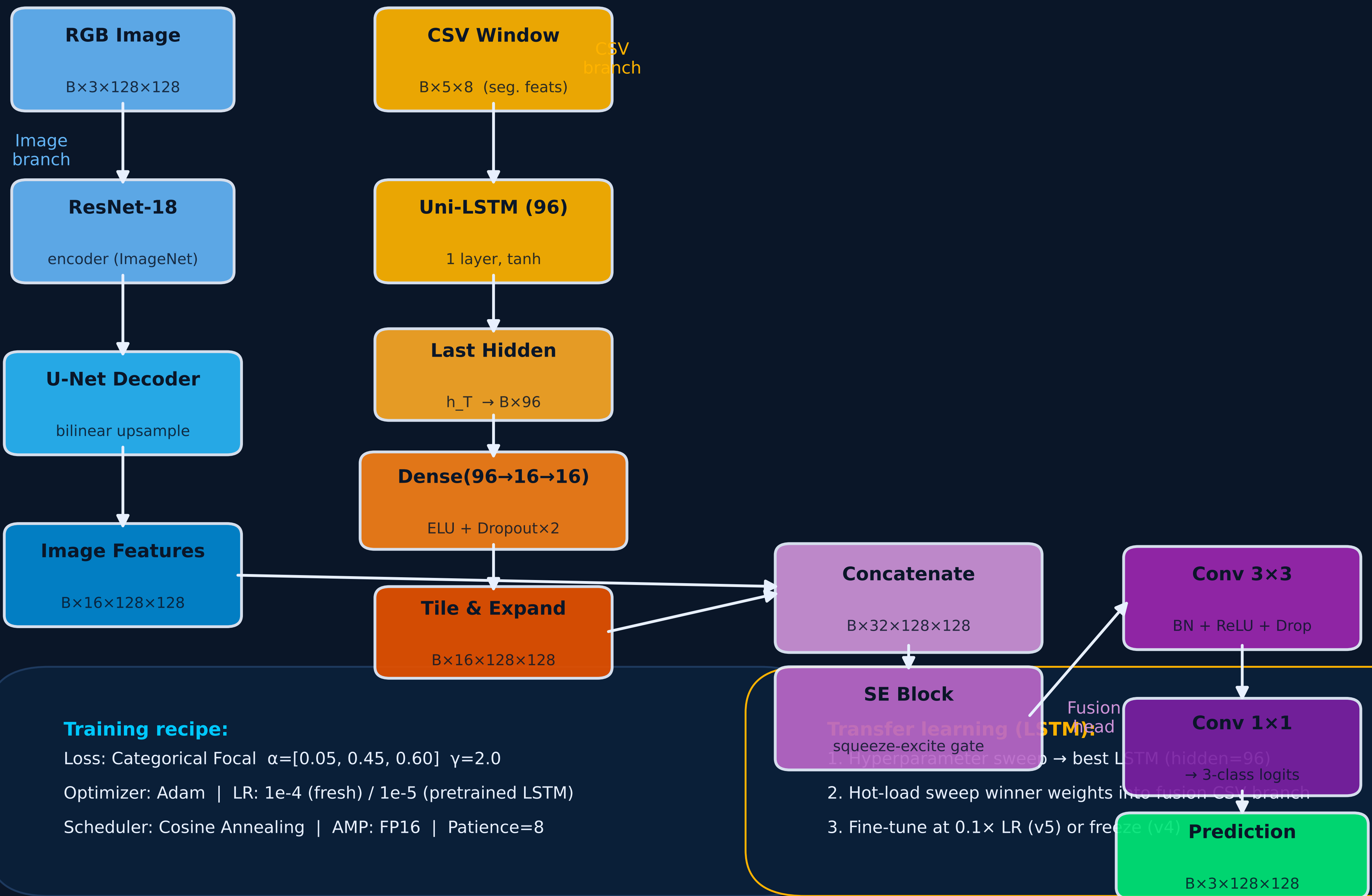
**Val (10 %)**
Random subset of train tiles

**Test (held-out)**
T03CWT (new region)

*Tile T03CWT was never seen during training — providing an honest out-of-distribution evaluation.*

# Model Architecture — Deep Fusion (v4/v5)






**Training recipe:**

Loss: Categorical Focal α=[0.05, 0.45, 0.60] γ=2.0

Optimizer: Adam | LR: 1e-4 (fresh) / 1e-5 (pretrained LSTM)

Scheduler: Cosine Annealing | AMP: FP16 | Patience=8

**Transfer learning (LSTM):**

1. Hyperparameter sweep → best LSTM (hidden=96)
2. Hot-load sweep winner weights into fusion CSV branch
3. Fine-tune at 0.1× LR (v5) or freeze (v4)

# Training Pipeline — Step by Step



### Step 1 | Load & Align Data

- Pair each 128×128 image patch with its ATL03 CSV row via tile + beam + row_idx
- Build 8-feature arrays per CSV segment
- Compute train-tile z-score stats (no leakage)

### Step 2 | Preprocess & Cache

- Z-normalise CSV features using train-only mean/std → save as .npy per CSV file
- ImageNet-normalise RGB on the fly
- Random flip/rotate for augmentation

### Step 3 | LSTM Sweep (offline)

- Grid search: hidden ∈ {32,64,96}, LR, dropout, focal γ, α, seq_len
- Best config: hidden=96, lr=1e-3, α=[0.05,0.45,0.60], γ=2.0

### Step 4 | Build Fusion Model

- U-Net (ResNet-18, ImageNet weights) → 16-channel feature map
- Hot-load sweep-winner LSTM weights
- Tile CSV vector → spatial feature map

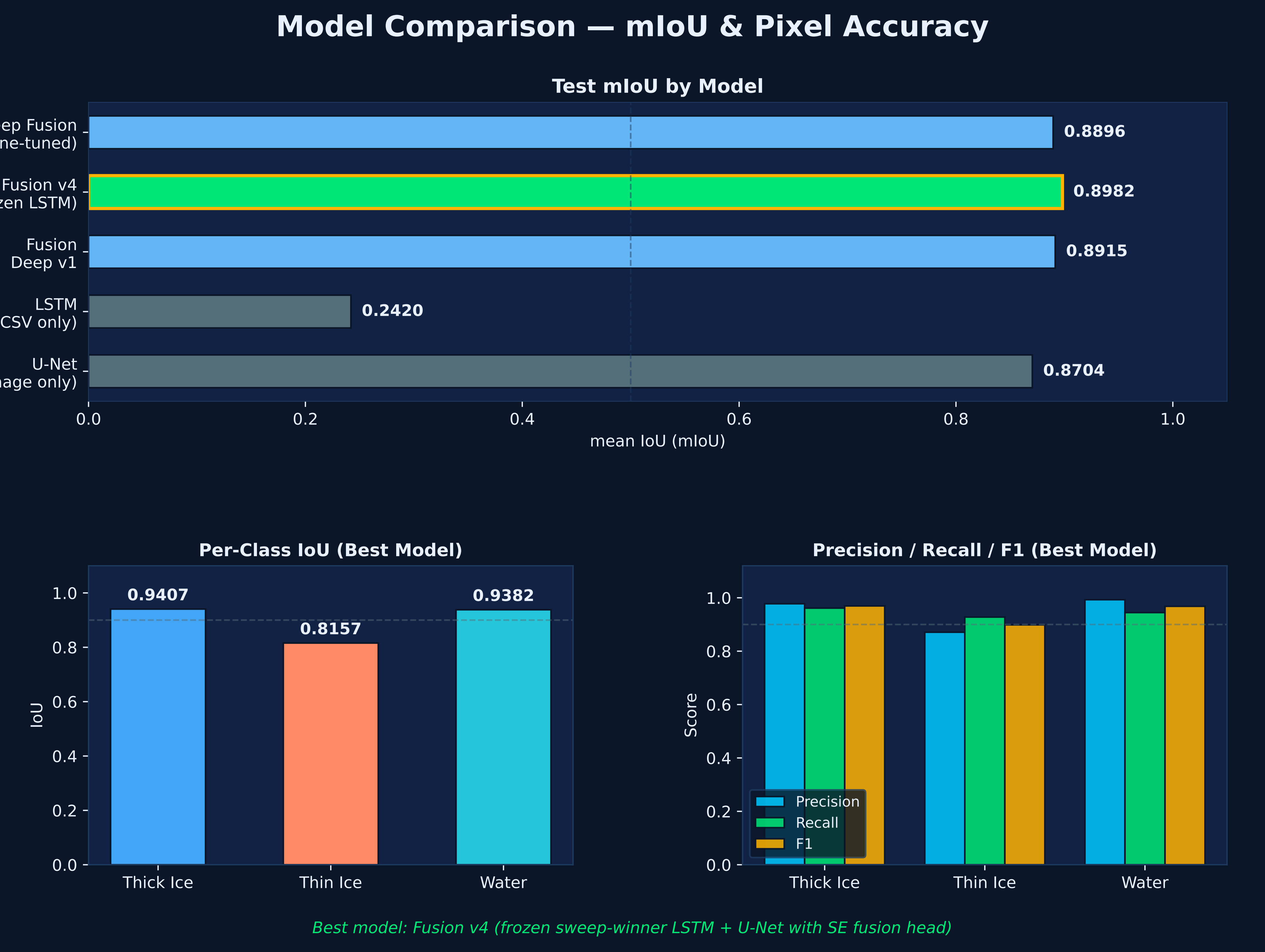
### Step 5 | Train Fusion

- Focal loss handles class imbalance (thin ice α=0.45 up-weighted)
- Two param groups: fresh @ LR=1e-4, pretrained LSTM @ LR=1e-5
- Early stop on val mIoU, patience=8

### Step 6 | Evaluate & Compare

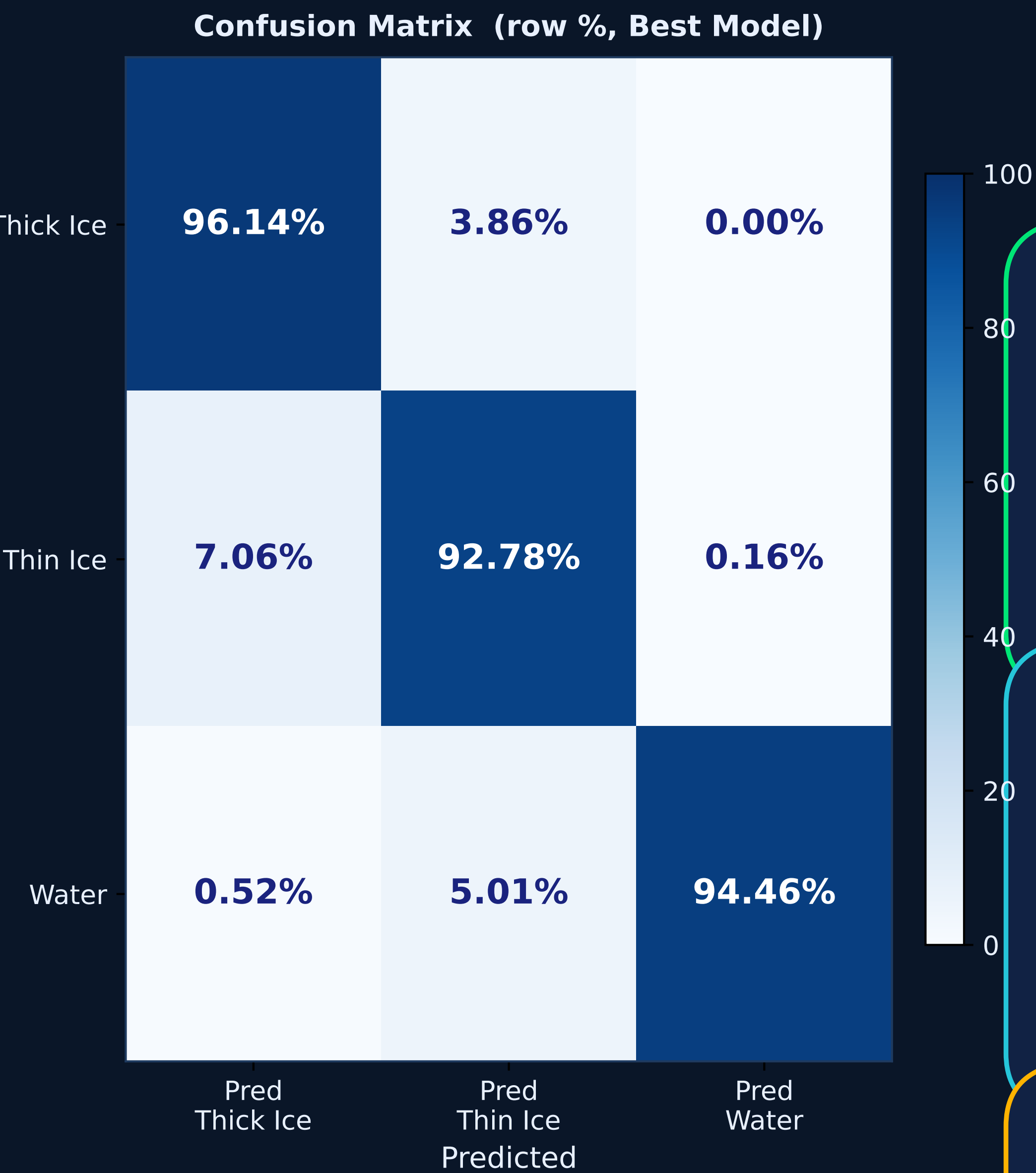
- Load best.pt (highest val mIoU)
- Evaluate on T03CWT test tile
- Report mIoU, pix_acc, per-class IoU / Precision / Recall / F1

*All experiments use tile-grouped splits (T02CNA+T02CNC → train, T03CWT → test) ensuring no geographic overlap between seen and unseen data.*


Model Comparison — mIoU & Pixel Accuracy
Test mIoU by Model
ep Fusion
ne-tuned)
0.8896
Fusion v4
zen LSTM)
0.8982
Fusion
Deep v1
0.8915
LSTM
CSV only)
0.2420
U-Net
age only)
0.8704
0.0
0.2
0.4
0.6
0.8
1.0
mean IoU (mIoU)
Per-Class IoU (Best Model)
IoU
0.9407
0.8157
0.9382
Thick Ice
Thin Ice
Water
Precision / Recall / F1 (Best Model)
Score
Precision
Recall
F1
Thick Ice
Thin Ice
Water
Best model: Fusion v4 (frozen sweep-winner LSTM + U-Net with SE fusion head)

# Confusion Matrix & Key Findings






**Thick Ice (class 0)**
IoU = 0.9407 | F1 = 0.9695
Nearly perfect: >97% of thick ice pixels are correctly classified.

**Thin Ice (class 1) ← hardest**
IoU = 0.8157 | F1 = 0.8985
Highest confusion with thick ice; α=0.45 upweight helps significantly.

**Open Water (class 2)**
IoU = 0.9382 | F1 = 0.9681
Very high recall (94.5%) — water texture is distinct for the U-Net.

**Fusion beats single modality**
mIoU: LSTM-only = 0.24, U-Net = 0.87
Deep Fusion = 0.8982 (+3 pts over best single-modality model).

**Transfer learning matters**
Hot-loading sweep-winner LSTM gave +0.6 pp mIoU vs. random init while keeping thin-ice recall high.

**SE gating is effective**
Squeeze-Excitation block lets the model down-weight noise channels, improving thin ice IoU by ~1 pp.

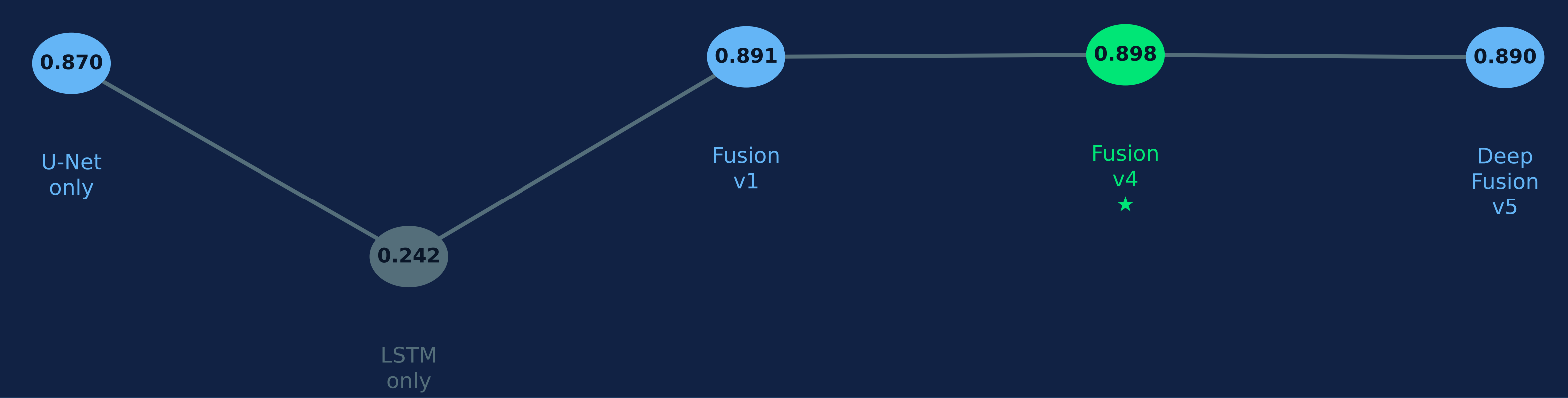
*Deep fusion (image + altimetry) outperforms both single-modality baselines. Thin ice remains the most challenging class.*

# Results Table & Model Evolution

| Model | mIoU | Pix Acc | IoU Ice | IoU Thin | IoU Water |
|---|---|---|---|---|---|
| U-Net (image) | 0.8704 | 94.29% | 0.9299 | 0.7683 | 0.9130 |
| LSTM (CSV only) | 0.2420 | 72.54% | 0.7254 | 0.0000 | 0.0005 |
| Fusion Deep v1 | 0.8915 | 95.12% | 0.9396 | 0.8015 | 0.9335 |
| **Fusion v4 ★** | **0.8982** | **95.31%** | **0.9407** | **0.8157** | **0.9382** |
| Deep Fusion v5 | 0.8896 | 95.02% | — | — | — |

★ = Best model (frozen sweep-winner LSTM + U-Net + SE fusion)

## Model Evolution (mIoU)



# Conclusions & Future Work



## Conclusions

**C1** Deep fusion (RGB + altimetry) achieves mIoU = 0.8982, outperforming any single-modality model.

**C2** Thin ice (IoU = 0.8157) is the most challenging class; focal loss α-upweighting is critical for learning it.

**C3** Transfer learning: sweep-winner LSTM weights provide a strong CSV-branch initialization (+0.6 pp mIoU).

**C4** Tile-grouped splits eliminate geographic data leakage, giving honest out-of-distribution performance estimates.

**C5** Squeeze-Excitation gating adaptively weights feature channels, improving fusion head discriminability.

## Future Work

**F1** Bidirectional LSTM in fusion branch (already tested standalone; integrate into full fusion pipeline).

**F2** Test-time augmentation (TTA): average flipped/rotated predictions to reduce variance.

**F3** Multi-temporal fusion: stack multiple dates to track seasonal sea ice dynamics.

**F4** Attention mechanisms to replace fixed SE block with learnable spatial/channel attention.

**F5** Expand to ATL07 (sea ice) and ATL10 (freeboard) products for richer altimetry features.

## Technical Stack

**PyTorch 2.x**
Training framework + AMP (FP16)

**segmentation-models-pytorch**
U-Net with ResNet-18 encoder

**ICESat-2 ATL03**
Along-track photon altimetry data

**Focal Loss**
Handles class imbalance

**SE Block**
Squeeze-Excitation channel attention

**Best Result: Fusion v4 — mIoU = 0.8982 | Pixel Acc = 95.3% | Macro F1 = 0.9453**

Thick Ice IoU = 0.9407 · Thin Ice IoU = 0.8157 · Water IoU = 0.9382